

The empirical recurrence for OEIS A231954: recovering a conjecture that is not written down

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Abstract

OEIS A231954 records “Empirical recurrence of order 50”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 67$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 50, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 50 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A231954 is “Number of $(n+1) \times (5+1)$ 0..1 arrays with every element equal to some horizontal, vertical, diagonal or antidiagonal neighbor, with top left element zero.”. It has offset 1 and begins

888, 58350, 3292876, 193955142, 11501370840, 681107245458, 40322931224352, 2387295795105696, ...

The entry states:

Empirical recurrence of order 50 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:45:16, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4160$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 67$ of the 4160, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 67$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 8320$ exact terms of the model returns order 50 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{50} c_i a(n-i),$$

c_1	54	c_{14}	−282761281025	c_{27}	39554088713846	c_{40}	−3758893975912
c_2	154	c_{15}	−501606600010	c_{28}	57574952854905	c_{41}	−1443574325872
c_3	8030	c_{16}	1586407448471	c_{29}	42100047370211	c_{42}	−89179499584
c_4	62015	c_{17}	4063352889818	c_{30}	−8182942696178	c_{43}	401115314752
c_5	161709	c_{18}	−3594635264566	c_{31}	−53421832511311	c_{44}	326779533504
c_6	956998	c_{19}	−14816436116470	c_{32}	−43363874409525	c_{45}	189062005248
c_7	−7049971	c_{20}	−1047218518324	c_{33}	−15303137756040	c_{46}	76467139584
c_8	−86957814	c_{21}	23660856179781	c_{34}	11730330926531	c_{47}	24877025280
c_9	−137979376	c_{22}	19338697665583	c_{35}	27343984053440	c_{48}	6462581760
c_{10}	363942871	c_{23}	751482785045	c_{36}	17791297619006	c_{49}	1115873280
c_{11}	2945823663	c_{24}	−24344068686270	c_{37}	4543193094924	c_{50}	176947200
c_{12}	19226603421	c_{25}	−51121221055396	c_{38}	−3217566854948		
c_{13}	10414106157	c_{26}	−20482384909352	c_{39}	−5935130205648		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 50, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $50 + 4$ terms removed returns order 50 as well, so $\deg q_{\min} = 50 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $50 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 50 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A231954.
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- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.